OPERATING SYSTEMS

Process Management and Scheduling

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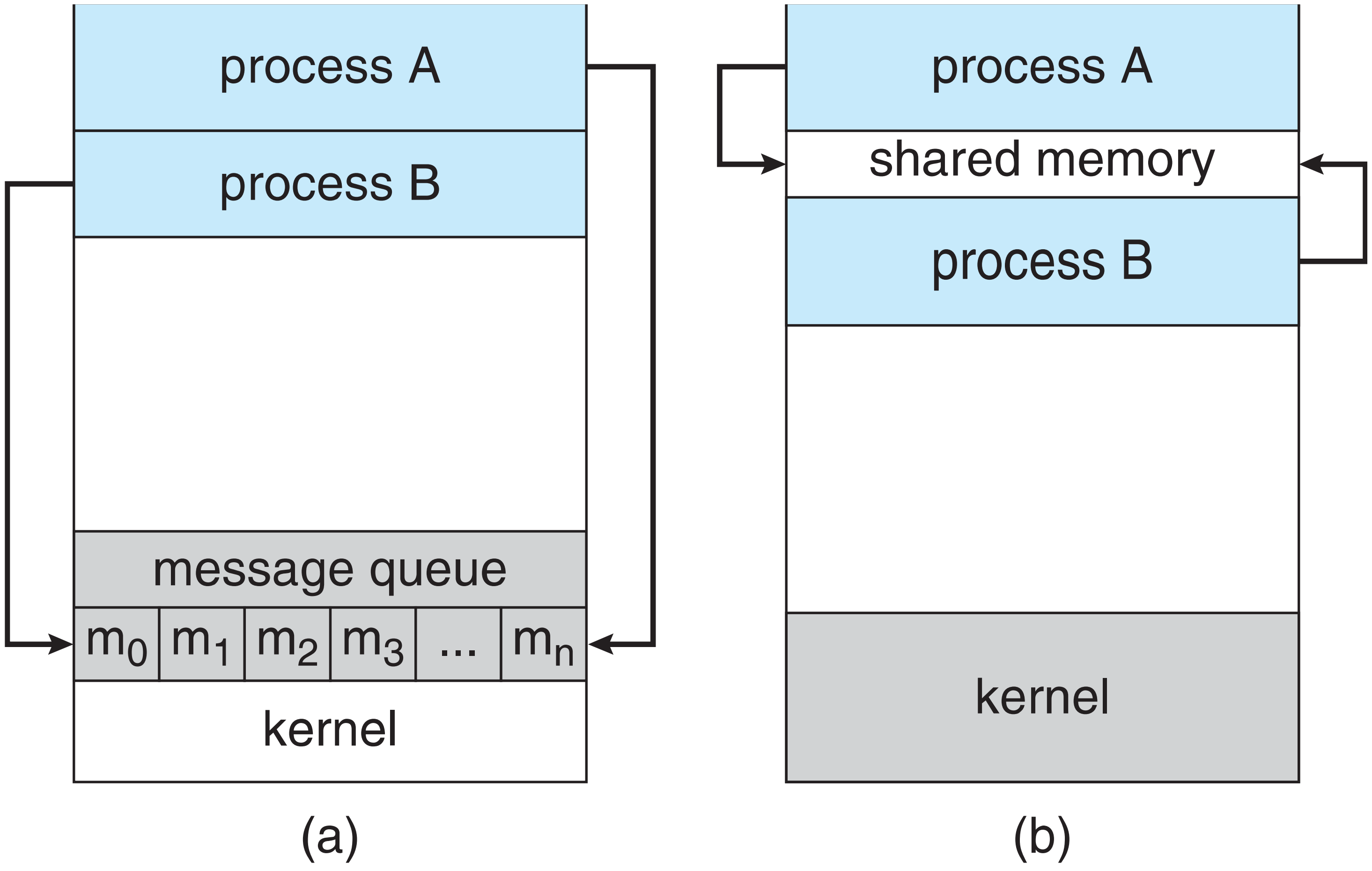
- Processes within a system may be ***independent*** or ***cooperating***
  - When processes execute they produce some **computational results**
  - ***Independent*** process cannot affect (or be affected) by results of another process
  - ***Cooperating*** process can affect (or be affected) by results of another process
- Advantages of process cooperation
  - Information sharing
  - Computation speed-up
  - Modularity

Interprocess Communication

- For fast exchange of information, cooperating processes need some **interprocess communication** (**IPC**) **mechanisms**
- Two models of IPC
  - **Message passing**
  - **Shared memory**

Interprocess Communication



Shared Memory

- An area of memory is shared among the processes that wish to communicate
- The communication is under the control of the users processes, not the OS.
- Major issue is to provide mechanism that will allow the user processes to synchronize their actions when they access shared memory.

Message Passing- Buffering

- Messages will be queued up
- Queues can be implemented in one of three ways
  1. Zero capacity – no messages are queued on a link. Sender must wait for receiver
  2. Bounded capacity – finite length of *n* messages. Sender must wait if link full
  3. Unbounded capacity – infinite length. Sender never waits

Message Passing

- Mechanism for processes to communicate and to synchronize their actions
- Message system – processes communicate with each other without resorting to shared variables
- IPC facility provides two operations:
  - send(message)
  - receive(message)
- If P and Q wish to communicate, they need to:
  - Establish a communication link between them
  - Exchange messages via send/receive

Message Passing – Naming – Direct Communication

- Processes must name each other explicitly:
  - **`send`** (*P, message*) – send a message to process P
  - **`receive`**(*Q, message*) – receive a message from process Q

- Properties of a direct communication link
  - Links are established automatically
  - Between each pair there exists exactly one link
  - The link is usually bi-directional

Message Passing – Naming – Indirect Communication

- Messages are directed and received from mailboxes (also referred to as ports)
  - Each mailbox has a unique id
  - Processes can communicate only if they share a mailbox

- Properties of an indirect communication link
  - Link established only if both members of the pair have a shared mailbox
  - A link may be associated with many processes
  - Each pair of processes may share several communication links
  - Link may be unidirectional or bi-directional

Message Passing – Naming – Indirect Communication

- Operations
  - create a new mailbox (port)
  - send and receive messages through mailbox
  - destroy a mailbox

- Primitives are defined as:

  send(*A, message*) – send a message to mailbox A

  receive(*A, message*) – receive a message from mailbox A

THANK YOU

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